$\mathbb{F}_{p^3}$-Frobenius nonclassical curves $y^n = f(x)$ and their geometry

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Abstract

Frobenius nonclassical curves over finite fields are interesting objects, with remarkable applications through their connection with minimal value set polynomials. This work gives a complete characterization of all $\mathbb{F}_{p^3}$-Frobenius nonclassical curves of type $y^n = f(x)$, and explicitly determines their genus, automorphism group, and number of $\mathbb{F}_{p^3}$-rational points.

Key Words : Frobenius nonclassical curves, automorphism group, Weierstrass semigroup.

1 Introduction

An irreducible plane curve $\mathcal{X} \subseteq \mathbb{P}^2$ defined over a finite field $\mathbb{F}_q$ of order q is called $\mathbb{F}_q$-Frobenius nonclassical if the $\mathbb{F}_q$-Frobenius map takes each nonsingular $P \in \mathcal{X}$ to the tangent line to $\mathcal{X}$ at P. Stöhr and Voloch [SV86] proved a bound that suggests that Frobenius nonclassical curves tend to have many rational points, which is confirmed in [HV90] and [BH17].

In [Bor16], a connection is established between $\mathbb{F}_q$-Frobenius nonclassical curves and minimal value set polynomials (MVSP), i.e., nonconstant polynomials $f(X) \in \mathbb{F}_q[X]$ for which the value set $\mathrm{V}_f = \{f(\alpha) : \alpha \in \mathbb{F}_q\}$ has the smallest possible size $\left\lfloor \frac{q-1}{\deg(f)} \right\rfloor + 1$. In [BR21], the authors characterize

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all MVSPs in $\mathbb{F}_{p^3}[x]$ and prove that if $\mathcal{F} : y^n = f(x)$, with $1 < n < p^3 - 1$ is an $\mathbb{F}_{p^3}$-Frobenius nonclassical curve then $n = p^2 + p + 1$, and there are approximately p^8 possibilities for the polynomial $f(x) \in \mathbb{F}_{p^3}[x]$.

This work proves that this set of $\mathbb{F}_{p^3}$-Frobenius nonclassical curves can be essentially reduced to three nonisomorphic curves, and their primary birational invariants can be determined.

Theorem 1.1. *Let $\mathcal{F} : y^n = f(x)$ be an $\mathbb{F}_{p^3}$-Frobenius nonclassical curve with $1 < n < p^3 - 1$. Then $\mathcal{F}$ is projectively equivalent to one of:*

$$\mathcal{F}_0 : y^{p^2+p+1} = x^{p^2+p+1}+1, \quad \mathcal{F}_1 : y^{p^2+p+1} = x^{p^2}+x^p+x, \quad \mathcal{F}_2 : y^{p^2+p+1} = x^{p^2+1}+x^p.$$

Moreover, some properties of $\mathcal{F} \cong \mathcal{F}_i, i = 0, 1, 2,$ are summarized below.

	Genus $\mathfrak{g}$	Aut	*Sing. Pts*	K	$\#\mathcal{F}(\mathbb{F}_{p^3})$
$\mathcal{F}_0$	$\dfrac{p^4+2p^3-p}{2}$	$(C_{p^2+p+1} \times C_{p^2+p+1}) \rtimes \mathbf{S}_3$	*0*	$\mathbb{F}_{p^3}$	$p^5 + 1 - (p^2 + p^3)$
$\mathcal{F}_1$	$\dfrac{p^4+p^3-p^2-p}{2}$	$(C_p \times C_p) \rtimes C_{p^3-1}$	*1*	$\mathbb{F}_{p^3}$	$p^5 + 1$
$\mathcal{F}_2$	$\dfrac{p^4+p}{2}$	$C_{p^4+p^2+1} \rtimes C_2$	*2*	$\mathbb{F}_{p^6}$	$p^5 + 1 + (p^2 + p^3)$

where C_m denotes the cyclic group of order m, and $K \supseteq \mathbb{F}_{p^3}$ the smallest field over which $\mathcal{F}$ is isomorphic to $\mathcal{F}_i$.

Remark 1.2. The tabulated properties of $\mathcal{F}_0$ and $\mathcal{F}_1$, particularly their automorphism groups, are well-established results included here for completeness; see [Sti99] for $\mathcal{F}_0$ and [BMZ19] for $\mathcal{F}_1$. The determination of $\mathrm{Aut}(\mathcal{F}_2)$ constitutes an original contribution of this work.

2 Preliminaries

2.1 $\mathbb{F}_{p^3}$-Frobenius nonclassical curves

A necessary condition for the curve $\mathcal{F} : y^n = f(x)$ over $\mathbb{F}_{p^3}$ to be $\mathbb{F}_{p^3}$-Frobenius nonclassical is that $n \mid p^3 - 1$ [Bor16, Theorem 5.3].

Theorem 2.1 ([BR21], Theorem 4.2)**.** *An irreducible plane curve $\mathcal{F} : y^n = f(x)$, defined over $\mathbb{F}_{p^3}$, with $1 < n < p^3 - 1$, is $\mathbb{F}_{p^3}$-Frobenius nonclassical if and only if $n = p^2 + p + 1$, and*

$$f(x) = \alpha x^{p^2+p+1} + \beta x^{p^2+p} + \beta^p x^{p^2+1} + \beta^{p^2} x^{p+1} + \gamma^{p^2} x^{p^2} + \gamma^p x^p + \gamma x + \delta,$$

for some $\alpha, \delta \in \mathbb{F}_p$, $\beta, \gamma \in \mathbb{F}_{p^3}$.

Our first step is to show that, up to projective equivalence, the curves in Theorem 2.1 can be reduced to only three. We begin with the following.

Theorem 2.2. *If $\mathcal{F} : y^n = f(x)$ is an $\mathbb{F}_{p^3}$-Frobenius nonclassical curve, with $1 < n < p^3 - 1$, then $\mathcal{F}$ is $\mathbb{F}_{p^3}$-projectively equivalent to $y^{p^2+p+1} = ax^{p^2+p+1} + b(x^{p^2} + x^p + x) + c$, for some $a, b, c \in \mathbb{F}_p$.*

Proof. By Theorem 2.1 there exist $\alpha, \delta \in \mathbb{F}_p$ and $\beta, \gamma \in \mathbb{F}_{p^3}$ such that

$$f(x) = \alpha x^{p^2+p+1} + \beta x^{p^2+p} + \beta^p x^{p^2+1} + \beta^{p^2} x^{p+1} + \gamma^{p^2} x^{p^2} + \gamma^p x^p + \gamma x + \delta.$$

Note that we can assume $\beta = 0$. Indeed, if $\beta \neq 0$, then the transformation $x \mapsto x + \lambda$ can be applied to f, where $\lambda = -\beta/\alpha$, when $\alpha \neq 0$; otherwise, λ is the unique $\mathbb{F}_{p^3}$-root of the permutation polynomial $\beta^{p^2} x^p + \beta^p x^{p^2} + \gamma$. In the latter case, we additionally apply the change of projective coordinates $(x, y, z) \mapsto (z, y, x)$. Therefore, up to a projective transformation, $\mathcal{F}$ is given by $y^{p^2+p+1} = \alpha x^{p^2+p+1} + \gamma^{p^2} x^{p^2} + \gamma^p x^p + \gamma x + \delta$, with $\alpha, \delta \in \mathbb{F}_p$ and $\gamma \in \mathbb{F}_{p^3}$. To complete the proof, it suffices to note that if $\gamma \neq 0$, the transformation $x \mapsto \gamma^{p^2+p} x$ will leads us to the desired equation, with $a = \alpha\gamma^{2(p^2+p+1)}$, $b = \gamma^{p^2+p+1}$, and $c = \delta$. $\square$

The curves as in Theorem 3.2 are investigated in Section 3.

2.2 Weierstrass semigroups

Let $\mathcal{X}$ be a projective, nonsingular, and irreducible algebraic curve over $\mathbb{K}$, and P be a point of $\mathcal{X}$. A non-negative integer s is called a nongap at P if there exists a rational function $\varphi \in \mathbb{K}(\mathcal{X})$ with pole divisor $\operatorname{div}(\varphi)_\infty = sP$. The set $H(P)$ of all nongaps at P forms a numerical semigroup, which is referred to as the Weierstrass semigroup of $\mathcal{X}$ at P. For Kummer extensions $\mathbb{K}(\mathcal{X})/\mathbb{K}(x)$ of the shape $y^n = f(x)$, a characterization of the nongaps at totally ramified points has been provided in [ABQ18]; this characterization is a fundamental tool for the results below.

3 Characterization of the curves

By [Bor16, Theorem 5.3 and 5.6], an $\mathbb{F}_8$-Frobenius nonclassical curve of type $y^n = f(x)$, with $n > 1$ is $\mathbb{F}_8$-projectively equivalent to $\mathcal{F}_i : y^{p^2+p+1} = f_i(x)$,

$$f_i(x) = \begin{cases} x^7 + 1 & \text{if } i = 0, \\ x^4 + x^2 + x & \text{if } i = 1, \\ x^7 + x^4 + x^2 + x + 1 & \text{if } i = 2. \end{cases}$$

One can easily check that $\mathcal{F}_2$ is projectively equivalent to $y^7 = x^5 + x^2$ over $\mathbb{F}_{64}$, but not over $\mathbb{F}_8$, as they have a different a number of points in $\mathbb{P}^2(\mathbb{F}_8)$. We will show that this is a general fact: up to projective equivalence, there are only three $\mathbb{F}_{p^3}$-Frobenius nonclassical curves $y^n = f(x)$, with $1 < n < p^3 - 1$.

We begin with a technical lemma, the proof of which is omitted here.

Lemma 3.1. *Let p be an odd prime, $b, c \in \mathbb{F}_p$, $g_1(x) = cg_2(x) + 2g_3(x)$ where*

$$g_2(x) = x^{p^2+p+1} + b(x^{p^2} + x^p + x) + c, \quad g_3(x) = -cx^{p^2+p+1} + b^2(x^{p^2+p} + x^{p^2+1} + x^{p+1}) + b^3.$$

If $c^2 + 4b^3$ is not a square in $\mathbb{F}_p$, then $\gcd(g_1(x), g_2(x)) = (bx^2 + cx - b^2)^p$, and $g_i(x)/(bx^2 + cx - b)^p$ is a factor of $x^{p^3} - x$, for $i = 1, 2$.

Proof. (Sketch) Let t be a multiple root of $g_i(x)$. Then t and $t^p \neq t$ are the roots of $bx^2 + cx - b^2$; also, every common root of $g_1(x)$ and $g_2(x)$ is t or t^p. By [Bor16, Corollary 5.4], t has multiplicity a multiple of p for $g_i(x)$, and the other roots $\neq t, t^p$ of $g_i(x)$ are simple and in $\mathbb{F}_{p^3}$. But (direct computation) $g_i'(x)$ is the p-power of a separable polynomial. The claim follows. $\square$

Theorem 3.2. *Let p be an odd prime, and let $\mathcal{F} : y^n = f(x)$ be an $\mathbb{F}_{p^3}$-Frobenius nonclassical curve, with $1 < n < p^3 - 1$. If $\lambda \in \mathbb{F}_p$ is not a square, then $\mathcal{F}$ is $\mathbb{F}_{p^3}$-projectively equivalent to $\mathcal{F}_i : y^{p^2+p+1} = f_i(x)$, where*

$$f_i(x) = \begin{cases} x^{p^2+p+1} + 1 & \text{if } i = 0, \\ x^{p^2} + x^p + x & \text{if } i = 1, \\ x^{p^2+p+1} + \lambda(x^{p^2} + x^p + x) & \text{if } i = 2. \end{cases}$$

Moreover, $\mathcal{F}_2$ is $\mathbb{F}_{p^6}$-projectively equivalent to $y^{p^2+p+1} = x^{p^2+1} + x^p$.

Proof. By Theorem 2.2, $\mathcal{F} : y^{p^2+p+1} = \alpha x^{p^2+p+1} + \beta(x^{p^2} + x^p + x) + \gamma$ with $\alpha, \beta, \gamma \in \mathbb{F}_p$. $\mathcal{F}_0$ and $\mathcal{F}_1$ occurs when $\beta = 0$ and $\alpha = 0$ respectively, through

$$(x, y) \mapsto \begin{cases} (\vartheta x, \varepsilon y) & \text{if } \beta = 0, \\ (x/\beta + \delta, y) & \text{if } \alpha = 0, \end{cases} \tag{3.1}$$

where $\vartheta, \varepsilon, \delta \in \mathbb{F}_{p^3}$ satisfy $\vartheta^{p^2+p+1} = \gamma/\alpha$, $\varepsilon^{p^2+p+1} = \gamma$ and $\delta^{p^2} + \delta^p + \delta = -\gamma/\beta$. If $\alpha\beta \neq 0$, the projectivity $y \mapsto \eta y$, with $\eta \in \mathbb{F}_{p^3}$ and $\eta^{p^2+p+1} = \alpha$, leads us to $y^{p^2+p+1} = x^{p^2+p+1} + b(x^{p^2} + x^p + x) + c$, with $b = \beta/\alpha \in \mathbb{F}_p^*$ and $c = \gamma/\alpha \in \mathbb{F}_p$. Now consider the following.

- Suppose that $c^2 + 4b^3$ is a square in $\mathbb{F}_p^*$. Let $u \in \mathbb{F}_p$ with $bu^2 + cu - b^2 = 0$. If $v \in \mathbb{F}_{p^3}$ is such that $v^{p^2+p+1} = (u^3 + bu)/b^3$, then the projectivity $(x, y, z) \mapsto \big(ux/b + z, vy, x - uz\big)$ maps $\mathcal{F}$ to $y^{p^2+p+1} = x^{p^2+p+1} + u/b^2$, which is equivalent to $\mathcal{F}_0$ by (3.1).
- Suppose that $c^2 + 4b^3 = 0$. Let $u \in \mathbb{F}_p$ with $bu^2 + cu - b^2 = 0$. Then $u^3 + 3bu + c = 0$ and $u^2 = -b$. The projectivity $(x, y, z) \mapsto \big(ux + z, y, x\big)$ maps $y^{p^2+p+1} = u(x^{p^2} + x^p + x) + 1$ to $\mathcal{F}$. Thus $\mathcal{F}$ is equivalent to $\mathcal{F}_1$ by (3.1).

- Suppose that c^2+4b^3 is not a square in $\mathbb{F}_p$. Thus there exists $v \in \mathbb{F}_p$ with $v^2 = \lambda/(c^2+4b^3)$, as $\lambda/(c^2+4b^3)$ is a nonzero square in $\mathbb{F}_p$. Let $t, s \in \mathbb{F}_{p^3}$ be such that $g_1(t) = 0$ and $s^{p^2+p+1} = g_2(t) \in \mathbb{F}_p^*$, with the $g_i(x)$'s as in Lemma 3.1. Then the projectivity $(x, y, z) \mapsto \big(tx+v(2b^2-ct)z, sy, x+v(2bt+c)z\big)$ maps $y^{p^2+p+1} = x^{p^2+p+1} + \lambda(x^{p^2}+x^p+x)$ to $\mathcal{F}$.

Finally, if $u, v, t \in \mathbb{F}_{p^6}$ are such that $u^2 = \lambda$, $v^{p^2+p+1} = -2u$ and $t^{p^4+p^2+1} = u^p$, then the projectivity $(x, y, z) \mapsto \big(t(x+uz), vy, t^{p^3}(-x+uz)\big)$ completes the proof. $\square$

Remark 3.3. By Theorem 3.2, all nonsingular $\mathbb{F}_{p^3}$-Frobenius nonclassical curves of the form $\mathcal{F} : y^n = f(x)$, with $1 < n < p^3-1$, are $\mathbb{F}_{p^3}$-projectively equivalent to the Fermat curve in the case $i = 0$. The case $i = 1$ corresponds to the Norm-Trace curve, which has a single singularity at its unique point at infinity. Meanwhile, the third curve, case $i = 2$, has two singularities. In either case, the singularities are unibranched [Sti09, Proposition 6.3.1].

Using Lemma 3.1 above, the following result can be proved (we omit the proof here). It gives the number of $\mathbb{F}_{p^3}$-rational points on the nonsingular model of $\mathcal{F}$ when it is projectively equivalent to $\mathcal{F}_i$. Note that this result can be easily verified in MAGMA when $p = 2$.

Proposition 3.4. *If N_i is the number of $\mathbb{F}_{p^3}$-rational points on the nonsingular model of $\mathcal{F}$ when it is projectively equivalent to $\mathcal{F}_i$, then*

$$N_i = \begin{cases} p^5-p^3-p^2+1 & \text{if } i=0 \\ p^5+1 & \text{if } i=1 \\ p^5+p^3+p^2+1 & \text{if } i=2. \end{cases}$$

Remark 3.5. It follows from Theorem 3.2 that $\mathcal{F}_2$ is $\mathbb{F}_{p^6}$-projectively equivalent to $y^{p^2+p+1} = x^{p^2+1}+x^p$. Since our next objectives involve the birational invariants of the curves $\mathcal{F}_i$, we will use the simplified model $y^{p^2+p+1} = x^{p^2+1}+x^p$ for the curve $\mathcal{F}_2$. This model has the advantage of not requiring the separation of the case $p = 2$.

Proposition 3.6. *The genus of $\mathcal{F}_i$ is*

$$\mathfrak{g}(\mathcal{F}_i) = \begin{cases} (p^4+2p^3-p)/2 & \text{if } i=0 \\ p(p+1)(p^2-1)/2 & \text{if } i=1 \\ (p^4+p)/2 & \text{if } i=2. \end{cases}$$

Proof. It follows from the factorization of $f_i(x)$ and [Sti09, Prop. 6.3.1]. $\square$

The tables below summarize genus and automorphisms of $\mathcal{F}_0$ and $\mathcal{F}_1$.

Table 1: Some birational invariants of $\mathcal{F}_0$

$\mathfrak{g}(\mathcal{F}_0)$	$(p^4+2p^3-p)/2$	Proposition 3.6
$\mathrm{Aut}(\mathcal{F}_0)$	$(C_{p^2+p+1}\times C_{p^2+p+1})\rtimes \mathbf{S}_3$	[Leo96, Theorem 1 and Comments on Page 257]

Table 2: Some birational invariants of $\mathcal{F}_1$

$\mathfrak{g}(\mathcal{F}_1)$	$p(p+1)(p^2-1)/2$	Proposition 3.6
$\mathrm{Aut}(\mathcal{F}_1)$	$(C_p\times C_p)\rtimes C_{p^3-1}$	[**?** , Theorem 3.1]

4 The curve $\mathcal{F}_2 : y^{p^2+p+1} = x^{p^2+1} + x^p$

Let $P_r = (\alpha_r, 0) \in \mathcal{F}_2$ where α_r is a root of $f(x) = x^{p^2+1} + x^p$, with $r \in [\![0, p^2-p+1]\!]$ and $\alpha_0 = 0$, and let $P_\infty = (1, 0, 0)$ be the unique point at infinity on $\mathcal{F}_2$. Note that P_0 and P_∞ are the unique singular points on $\mathcal{F}_2$.

4.1 Weierstrass semigroup

The points $P_r = (\alpha_r, 0)$, with $r \in [\![0, p^2-p+1]\!]$ and $\alpha_0 = 0$, and P_∞ are totally ramified by [Sti09, Proposition 6.3.1].

Lemma 4.1. *For $t \in [\![0, p-1]\!]$ the following holds:*

i) *$t(p^3-p-2)+p^2+p$ and p^2+p+1 are nongaps at P_r, for all $r \in [\![1, p^2-p+1]\!]$.*

ii) *tp^3+p^2+1 and p^2+p+1 are nongaps at P_0 and P_∞.*

Proof. We omit the details of the proof; it relies on [ABQ18, Corollary 3.6]. □

Theorem 4.2. *If $H(P_r)$ is the Weierstrass semigroup of $\mathcal{F}_2$ at P_r, then*

i) *$H(P_r) = \langle p^2+p+1, t(p^3-p-2)+p^2+p \mid t \in [\![0, p-1]\!]\rangle$, for $r \in [\![1, p^2-p+1]\!]$.*

ii) *$H(P_0) = H(P_\infty) = \langle p^2+p+1, tp^3+p^2+1 \mid t \in [\![0, p-1]\!]\rangle$.*

Proof. i) Let $H = \langle p^2+p+1, t(p^3-p-2)+p^2+p \mid t \in [\![0, p-1]\!]\rangle$. It follows from Lemma 4.1 that $H \subset H(P_r)$.

Let $H' = \langle p^2+1, p^2+p+1\rangle$ and

$$A = \{t(p^3-p-2)+u(p^2+p)+v(p^2+p+1) \mid t \in [\![1,p]\!],\ u \in [\![1,(p+1)(p-t)]\!] \text{ and } v \in [\![0,1]\!]\}.$$

If $u \geq (p+1)(p-t)$, then $u = (p+1)(p-t) + l$, for some $l \in \mathbb{N}$, and

$$t(p^3-p-2)+u(p^2+p)+v(p^2+p+1) = (p^2+p-2t+v)(p^2+p+1)+l(p^2+p) \in H'. \tag{4.1}$$

By some arithmetic computation, it can be proved that $H = H' \cup A$; we omit the details. Now, note that the numerical semigroup H' is telescopic. As H' has genus $\#(\mathbb{N} \setminus H') = (p^4 + 2p^3 - p)/2$, we have

$$\#(\mathbb{N} \setminus H) = \frac{p^4 + 2p^3 - p}{2} - (p^3 - p) = \frac{p^4 + p}{2} = \#(\mathbb{N} \setminus H(P_j)).$$

Therefore, $H(P_r) = H$.

ii) Let $H = \langle p^2 + p + 1, tp^3 + p^2 + 1 \mid t \in [\![0, p-1]\!]\rangle$ and $P \in \{P_0, P_\infty\}$. As before, one can check that $H \subset H(P)$, and $H = H' \cup A$, where the sets $H' = \langle p^2 + 1, p^2 + p + 1\rangle$ and

$$A = \{tp^3 + u(p^2 + 1) \mid t \in [\![1, p-1]\!],\ u \in [\![1, (p+1)(p-t)]\!]\},$$

are disjoint. Since H' is telescopic, $\#(\mathbb{N} \setminus H') = (p^4 + p^3)/2$ and

$$\#(\mathbb{N} \setminus H) = \frac{p^4 + p^3}{2} - \sum_{t=1}^{p-1}(p+1)(p-t) = \frac{p^4 + p}{2} = \#(\mathbb{N} \setminus H(P)).$$

Therefore, $H(P) = H$.

□

4.2 The automorphism group

Fix a primitive $(q^4 + q^2 + 1)$-primitive root of unity $\lambda \in \mathbb{F}_{q^6}$, and define

$$\sigma_\lambda : (x, y) \mapsto (\lambda^{q^2+q+1}x, \lambda^q y), \quad \tau : (x, y) \mapsto (1/x, y/x), \quad G := \langle \sigma_\lambda, \tau\rangle \leq \mathrm{PGL}(3, q^6).$$

Then $G \cong C_{q^4+q^2+1} \rtimes C_2$, and by direct computation $G \leq A := \mathrm{Aut}(\mathcal{F}_2)$.

Theorem 4.3. $\mathrm{Aut}(\mathcal{F}_2) = G$.

This section is devoted to the proof of Theorem 4.3. We will distinguish the arguments according to the parity of q only in the last part, when the geometry of the group depends on the field characteristic being odd or even.

It is easily checked that $O_1^G := \{P_0, P_\infty\}$ and $O_2^G := \{P_r = (\alpha_r, 0) \colon r = 1, \ldots, q^2 - q + 1\}$ are two short orbits under G, of size respectively 2 and $q^2 - q + 1$. The stabilizer in O_1^G is $G_{P_0} = G_{P_\infty} = \langle\sigma_\lambda\rangle \cong C_{q^4+q^2+1}$, while the stabilizer of a point in O_2^G is $\langle\sigma_\lambda^{q^2-q+1}, \sigma_\lambda^i\tau\rangle \cong C_{2(q^2+q+1)}$ for some involution $\sigma^i\tau \in G$; for instance, the stabilizer of $(-1, 0)$ is $\langle\sigma_\lambda^{q^2-q+1}, \tau\rangle$.

For $i = 1, 2$, let O_i^A be the A-orbit containing the G-orbit O_i^G.

Remark 4.4. From Theorem 4.2 it follows that O_1^A and O_2^A are distinct, because they contain points with different Weierstrass semigroups.

Lemma 4.5. $A_{P_0} = A_{P_\infty} = \langle\sigma_\lambda\rangle \cong C_{q^4+q^2+1}$.

Proof. We can write $A_{P_0} = S \rtimes C$, where S is the Sylow p-subgroup of A_{P_0} and C is a cyclic group of order coprime with p containing $\langle\sigma_\lambda\rangle$ (up to conjugacy). The group C acts on the set O_1^G of fixed points of its normal subgroup $\langle\sigma_\lambda\rangle$; thus C fixes both P_0 and P_∞. Therefore C maps x to a scalar multiple of x, and y to a scalar multiple of y; by direct checking using the equation of $\mathcal{F}_2$, this implies $C = \langle\sigma_\lambda\rangle$. Now suppose that $|S| > 1$, and let $\varphi \in S \setminus \{id\}$. As a nontrivial p-element, φ cannot map x to a multiple of x and y to a multiple of y; thus, $\varphi(P_\infty) \neq P_\infty$. This implies that φ does not commute with any nontrivial element $\sigma_\lambda^i \in \langle\sigma_\lambda\rangle$: otherwise, φ acts on the set of fixed points of σ_λ^i, which is either $O_1^G \cup O_2^G$ or O_1^G (according to i being or not a multiple of $q^2 - q + 1$), and in both cases we have $\varphi(P_\infty) = P_\infty$ by Remark 4.4, a contradiction. Therefore the action by conjugation of $\langle\sigma_\lambda\rangle$ on $S \setminus \{id\}$ is semiregular; in particular, $|S|$ is at least $1 + |\langle\sigma_\lambda\rangle| = q^4 + q^2 + 2$, whence $|S| > 2\mathfrak{g}(\mathcal{F}_2) + 1$. By [HKT08, Theorem 11.140], this implies that either G fixes P_0; or $\mathcal{F}_2$ is a Hermitian, Suzuki or Ree curve. None of these cases happens, and we have a contradiction. Then S is trivial, and the claim is proved. □

Remark 4.6. It is easy to determine all the short orbits under G as follows, depending on the characteristic.

- If q is even, then O_1^G and O_2^G are the only short G-orbits.
- If q is odd, there are exactly three short G-orbits; namely, O_1^G which is a tame A-orbit, O_2^G, and
$$O_3^G := \left\{(\alpha,\beta)\colon \alpha^{q^2-q+1} = 1, \beta^{q^2+q+1} = 2\alpha^q\right\}.$$

By Lemma 4.5 and the orbit-stabilizer theorem, if $O_1^A = O_1^G$ then $|A| = |G|$ and Theorem 4.3 holds. Thus, we assume from now on that $O_1^G \subsetneq O_1^A$.

Since O_1^A is distinct from O_2^A, and when q is odd O_1^A is also distinct from O_3^A, we have that $O_1^A \setminus O_1^G$ is a union of long G-orbits, so that

$$|O_1^A| = |O_1^G| + k|G| = 2 + 2k(q^4 + q^2 + 1) \quad \text{for some} \quad k \geq 1.$$

Then, by the orbit-stabilizer theorem,

$$|A| = |A_{P_0}| \cdot |O_1^A| = (q^4 + q^2 + 1) \cdot [2 + 2k(q^4 + q^2 + 1)] > 84(\mathfrak{g}(\mathcal{F}_2) - 1).$$

Then, by [HKT08, Theorem 11.56], there are few possibilities for the structure of the short A-orbits; since there exists a tame short A-orbit O_1^A with

$|A_P| = q^4+q^2+1 > 2$ for $P \in O_1^A$, the only possibility is that there are exactly two short A-orbits, one of them is tame and the other is non-tame. Therefore, the unique tame short A-orbit is O_1^A (by Lemma 4.5); the unique non-tame short A-orbit is O_2^A containing O_2^G, and O_2^A contains also O_3^G for q odd (by Remark 4.4).

Fix $P_2 \in O_2^A$, in particular choose $P_2 = (-1,0)$, and write $A_{P_2} = \tilde{S} \rtimes \tilde{C}$, where $\tilde{S}$ is a nontrivial p-group and $\tilde{C}$ is cyclic of order coprime with p. For $p=2$, $\tilde{S}$ contains $\langle \tau \rangle \cong C_2$ and we can assume up to conjugacy that $\tilde{C}$ contains $\langle \sigma_\lambda^{q^2-q+1} \rangle \cong C_{q^2+q+1}$; for $p>2$, we can assume that $\tilde{C}$ contains $\langle \sigma_\lambda^{q^2-q+1}\tau \rangle \cong C_{2(q^2+q+1)}$.

Arguing as in the proof of Lemma 4.5 we obtain, from the normality of $\langle \sigma_\lambda^{q^2-q+1} \rangle$ in $\tilde{C}$ and Remark 4.4, that $\tilde{C}$ acts on $O_1^G = \{P_0, P_\infty\}$ and acts on O_2^G fixing P_2. Thus, by Lemma 4.5, the pointwise stabilizer $\tilde{C}_{(P_0,P_\infty)}$ of $\{P_0,P_\infty\}$ in $\tilde{C}$ fixes P_2 and is contained $\langle \sigma_\lambda \rangle$, whence $\tilde{C}_{(P_0,P_\infty)} = \langle \sigma_\lambda^{q^2-q+1} \rangle$. Suppose now that $\tilde{C} \neq \tilde{C}_{(P_0,P_\infty)}$; then $[\tilde{C} : \tilde{C}_{(P_0,P_\infty)}] = 2$. In particular, $p>2$ as $p \nmid |\tilde{C}|$, and $\tilde{C} = \langle \sigma_\lambda^{q^2-q+1}\tau \rangle \cong C_{2(q^2+q+1)}$. Therefore, we have shown that $\tilde{C}$ is contained in G and has order q^2+q+1 or $2(q^2+q+1)$ according to $p=2$ or $p>2$. Now we consider $\tilde{S}$, dealing with $p>2$ and $p=2$.

4.2.1 The case $p>2$

If a nontrivial element $\varphi \in \tilde{S}$ commutes with a nontrivial element $\sigma \in \langle \sigma_\lambda^{q^2-q+1} \rangle \leq \tilde{C}$, then φ acts on the fixed points $O_1^G \cup O_2^G$, and hence acts on $O_1^G = \{P_0, P_\infty\}$. Then either φ fixes both P_0 and P_∞, which is not possible by Lemma 4.5, or φ swaps P_0 and P_∞, which is not possible since 2 does not divide the order of φ. Thus $\langle \sigma_\lambda^{q^2-q+1} \rangle$ acts semiregularly by conjugation on $\tilde{S} \setminus \{id\}$, so that q^2-q+1 divides $|\tilde{S}|-1$.

Remark 4.7. The following arithmetic fact is easy to prove: for any prime $\hat{p} \geq 2$ and any power $\hat{q} = \hat{p}^h$ of $\hat{p}$, if a power $\hat{p}^a$ of $\hat{p}$ satisfies $(\hat{q}^2+\hat{q}+1) \mid (\hat{p}^a-1)$, then $\hat{p}^a$ is indeed a power of $\hat{q}^3$. In fact, let $0 \leq r < 3h$ be the residue of a modulo $3h$. Then $(\hat{q}^2+\hat{q}+1) \mid (\hat{p}^r-1)$. If we had $r>0$, then $r = 2h+\varepsilon$ with $0<\varepsilon<h$, whence $(\hat{q}^2+\hat{q}+1) \mid (\hat{p}^{h+\varepsilon}+\hat{p}^\varepsilon)+1$, a contradiction.

By Remark 4.7, $\tilde{S}$ has size q^{3c} for some $c \geq 1$. We apply Hurwitz formula to the covering $\mathcal{F}_2 \to \overline{\mathcal{F}}_2 := \mathcal{F}_2/\tilde{S}$; Since $\tilde{S}$ is a p-group and fixes P_2, we have that the different exponent is at least $2(|\tilde{S}|-1)$, so that

$$2\mathfrak{g}(\mathcal{F}_2) - 2 \geq |\tilde{S}|(2\mathfrak{g}(\overline{\mathcal{F}}_2)-2) + 2(|\tilde{S}|-1),$$

whence

$$\mathfrak{g}(\overline{\mathcal{F}}_2) \leq \mathfrak{g}(\mathcal{F}_2)/|\tilde{S}| = \frac{q^4+q}{2} \cdot \frac{1}{|\tilde{S}|} \leq \frac{q^4+q}{2q^3}. \tag{4.2}$$

Since $\tilde{S}$ is normal in A_{P_2}, the quotient curve $\overline{\mathcal{F}}_2$ has a cyclic automorphims group $\overline{C} := A_{P_2}/\tilde{S}$ of order $2(q^2+q+1)$ which fixes the point $\overline{P}_2$ under P_2.

- Suppose $\mathfrak{g}(\overline{\mathcal{F}}_2) = 1$. Since $\overline{\mathcal{F}}_2$ is elliptic, the stabilizer of $\overline{P}_2$ has size at most 12 (see [HKT08, Theorem 11.94]), a contradiction to $|\overline{C}| > 12$.
- Suppose $\mathfrak{g}(\overline{\mathcal{F}}_2) \geq 2$. Then the abelian group $\overline{C}$ satisfies $2(q^2+q+1) = |\overline{C}| \leq 4\mathfrak{g}(\overline{\mathcal{F}}_2) + 4$ ([HKT08, Theorem 11.79]), a contradiction to (4.2).
- Finally, suppose $\mathfrak{g}(\overline{\mathcal{F}}_2) = 0$. Then $\mathrm{Aut}(\overline{\mathcal{F}}_2) \cong \mathrm{PGL}(2, \mathbb{K})$ and $\overline{C}$ is a finite cyclic subgroup of $\mathrm{PGL}(2, \mathbb{K})$ of order coprime with p. Therefore, $\overline{C}$ fixes two points $\overline{P}_2$ and $\overline{Q} \neq \overline{P}_2$, and $\overline{C}$ acts semiregularly on $\overline{\mathcal{F}}_2 \setminus \{\overline{P}_2, \overline{Q}\}$. Thus, that all fixed points of $\tilde{C}$ that are different from P_2 are in $\tilde{O}$; that is, $O_1^G \cup O_2^G \subseteq \tilde{O}$; a contradiction to Remark 4.4.

4.2.2 The case $p = 2$

If $|\tilde{S}| > 2$, i.e. $\langle \tau \rangle \subsetneq \tilde{S}$, then the strategy is the same as in the odd p case; we omit the details. Therefore, we assume from now on that $|\tilde{S}| = 2$, that is, $A_{P_2} = G_{P_2} \cong C_{2(q^2+q+1)}$. As we are assuming $G \subsetneq A$, the fact that $A_{P_2} = G_{P_2}$ implies that the A-orbit O_2^A is the union of O_2^G and $h \geq 1$ long G-orbits, so that $|O_2^A| = |O_2^G| + h|G| = q^2 - q + 1 + 2h(q^4 + q^2 + 1)$, $h \geq 1$.

We determine the contribution $i(\tau)$ of τ to the different exponent in $\mathcal{F}_2 \to \mathcal{F}_2/H$, for any group H containing τ; as $P_2 = (1, 0)$ is the only fixed point of τ, we have to count the number of higher ramification groups at P_2 containing τ. Consider the covering $\mathcal{F}_2 \to \mathcal{F}_2/A_{P_2}$; the quotient curve is rational, since A_{P_2} contains $\langle \sigma_\lambda^{q^2-q+1} \rangle$. The nontrivial elements of A_{P_2} are as follows: the unique p-element τ fixes only P_2; the $q^2 + q$ elements of order dividing $q^2 + q + 1$ fix exactly $|O_1^G \cup O_2^G| = q^2 - q + 3$ points and contribute $q^2 - q + 2$ to the different exponent; the remaining $q^2 + q$ elements fix only P_2 and contribute 1 to the different exponent. Thus, the Hurwitz formula reads

$$2\mathfrak{g}(\mathcal{F}_2) - 2 = |A_{P_2}|(2 \cdot 0 - 2) + i(\tau) + (q^2 + q)(q^2 - q + 3) + (q^2 + q) \cdot 1,$$

whence $i(\tau) = q^2 + q + 2$; this means that τ is in the j-th ramification group at P_2 if and only if $j \leq q^2 + q + 1$ (the 0-th being the stabilizer).

Now we apply the Hurwitz formula to the covering $\mathcal{F}_2 \to \mathcal{F}_2/A$, using what we obtained on the short $\mathrm{Aut}(\mathcal{F}_2)$-orbits. This yields

$$q^4 + q - 2 = -[2 + 2k(q^4 + q^2 + 1)] + [q^2 - q + 1 + 2h(q^4 + q^2 + 1)](q^2 + q);$$

whence, by direct computation, $k = h(q^2 + q)/2$. By the orbit-stabilizer theorem applied to both short orbits of A, we have $|A_{P_0}| \cdot |O_1^A| = |A_{P_2}| \cdot |O_2^A|$; using $k = h(q^2 + q)/2$, this yields an arithmetic contradiction to $h \geq 1$.

The proof of Theorem 4.3 is now complete.

References

[ABQ18] M. Abdón, H. Borges, and L. Quoos. Weierstrass points on kummer extensions. *Advances in Geometry*, 19(3):323–333, 2018.

[BH17] H. Borges and M. Homma. Points on singular frobenius nonclassical curves. *Bull Braz Math Soc*, New Series 48:93–101, 2017.

[BMZ19] M. Bonini, M. Montanucci, and G. Zini. On plane curves given by separated polynomials and their automorphisms. *Advances in Geometry*, 2019.

[Bor16] H. Borges. Frobenius nonclassical components of curves with separated variables. *Journal of Number Theory*, 159:402–425, 2016.

[BR21] H. Borges and L. Reis. Minimal value set polynomials over fields of size p^3. *Proc. Amer. Math. Soc.*, 149:3639–3649, 2021.

[HKT08] J. W. P. Hirschfeld, G. Korchmáros, and F. Torres. *Algebraic curves over a finite field*. Princeton Series in Applied Mathematics. Princeton University Press, Princeton, NJ, 2008.

[HV90] A. Hefez and J.F. Voloch. Frobenius non classical curves. *Arch. Math.*, 54:263–273, 1990.

[Leo96] H. W. Leopoldt. über die automorphismengruppe des fermatkörpers. *Journal of Number Theory*, 56:256–282, 1996.

[Sti99] H. Stichtenoth. The fermat curve in characteristic p. *Contemporary Mathematics*, 225:123–129, 1999.

[Sti09] H. Stichtenoth. *Algebraic Function Fields and Codes*. Springer-Verlag, 2 edition, 2009.

[SV86] K. O. Stöhr and J. F. Voloch. Weierstrass points and curves over finite fields. *Proc. London Math*, s3-52:1–19, 1986.